

Course Outcomes (COs)	Course Outcome Statements
CO1	Analyze the structure and interaction of computer system components.
CO2	Apply number system conversions and data representation techniques to solve computational problems.
CO3	Differentiate types of operating systems based on their functions and suitability for various applications.
CO4	Analyse computer network components, topologies, and data communication processes.
CO5	Identify appropriate emerging technologies to address the given engineering problems.

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
CO1	<i>TSO 1a.</i> Classify different types of computers based on application <i>TSO 1b.</i> Explain the organization of the CPU and its working process. <i>TSO 1c.</i> Illustrate memory architecture and hierarchy in a computer system. <i>TSO 1d.</i> Analyze the role of CPU registers and cache memory in system performance. <i>TSO 1e.</i> Identify motherboard components and their functions. <i>TSO 1f.</i> Explain the Von Neumann Architecture <i>TSO 1g.</i> Distinguish between machine language, assembly language, and high-level language. <i>TSO 1h.</i> Differentiate between compiler and interpreter.	Unit- 1.0: Fundamentals of Computer Systems (8hrs) 1.1 Classification of computers, Functional units 1.2 Central Processing Unit (CPU) organization and working 1.3 Memory: Architecture, Memory Representation, Memory Hierarchy 1.4 CPU Registers, Impact of Cache Memory on System Performance 1.5 Motherboard and its components 1.6 The Von-Neumann architecture 1.7 Machine language, assembly language, high-level programming languages Compiler, Interpreter
CO2	<i>TSO 2a.</i> Explain the fundamentals of data and its representation in computers.	Unit- 2.0: Data Representation (8hrs) 2.1 Data Representation & Number Systems. 2.2 Conversion between decimal and binary systems.

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	<i>TSO 2b.</i> Convert numbers between decimal and binary systems. <i>TSO 2c.</i> Convert numbers among binary, octal, and hexadecimal systems. <i>TSO 2d.</i> Perform binary arithmetic operations. <i>TSO 2e.</i> Differentiate signed number representations. <i>TSO 2f.</i> Perform signed number arithmetic. <i>TSO 2g.</i> Interpret character encoding schemes such as ASCII and Unicode. <i>TSO 2h.</i> Illustrate data representation in computer memory.	2.3 Conversion among binary, octal, and hexadecimal systems. 2.4 Binary arithmetic (addition, subtraction) 2.5 Signed number representation (1's complement). 2.6 Signed number representation (2's complement). 2.7 American Standard Code for Information Interchange (ASCII), Unicode Data representation in memory.
CO3	<i>TSO 3a.</i> Explain the concept and types of computer software. <i>TSO 3b.</i> Describe components of system software. <i>TSO 3c.</i> Classify application software based on usage. <i>TSO 3d.</i> Explain objectives of an operating system. <i>TSO 3e.</i> Compare operating system types for specific applications. <i>TSO 3f.</i> Analyze process and memory management techniques. <i>TSO 3g.</i> Evaluate user interfaces based on usability and efficiency. <i>TSO 3h.</i> Analyze file system organization for efficient data management. <i>TSO 3i.</i> Compare features of major operating systems.	Unit- 3.0: Computer Software and Operating Systems (9hrs) 3.1 Introduction to Software, Types of Software 3.2 System Software 3.3 Application Software 3.4 Operating System (OS), Objectives of Operating System 3.5 Types of OS (Batch, Time-sharing, Distributed, Real-time) 3.6 Functions of OS (Process, Memory Management) 3.7 User Interface (Command Line Interface Vs Graphical User Interface) 3.8 File system basics examples: Windows, Linux, macOS.

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CO4	<i>TSO 4a.</i> Explain the fundamental concepts of computer networking. <i>TSO 4b.</i> Justify the benefits of networking. <i>TSO 4c.</i> Explain the process and key components of data transmission in networks. <i>TSO 4d.</i> Differentiate between different data transmission media used in networking. <i>TSO 4e.</i> Analyze data transmission across different media. <i>TSO 4f.</i> Classify computer networks based on scale. <i>TSO 4g.</i> Differentiate between various network topologies. <i>TSO 4h.</i> Analyze the functions of network devices in data communication. <i>TSO 4i.</i> Describe IP addresses in a computer network <i>TSO 4j.</i> Explain basic communication protocols in networking. <i>TSO 4k.</i> Analyze Internet components and client-server architecture.	Unit- 4.0: Computer Networks and Internet Basics (9hrs) 4.1 Introduction to Computer Networks. 4.2 Importance of Networking. 4.3 Data Transmission Media, Data Transmission across Media. 4.4 Data Transmission in a Network. 4.5 Network Types: Local Area Network (LAN), Wide Area Network (WAN), Metropolitan Area Network (MAN). 4.6 Network Topologies (Star, Bus, Ring, Mesh), Communication Protocol. 4.7 Network Devices (Hub, Switch, Router, Modem, Gateway), Wireless Networking. 4.8 Introduction to Internet: Internet Protocol (IP) Address. 4.9 Transmission Control Protocol (TCP), User Datagram Protocol (UDP), Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Domain Name System (DNS), Uniform Resource Locator (URL), Client-Server Architecture.
CO5	<i>TSO 5a.</i> Explain the concept and architecture of IoT. <i>TSO 5b.</i> Analyze IoT applications in real-world scenarios. <i>TSO 5c.</i> Explain basic cybersecurity concepts and terminologies. <i>TSO 5d.</i> Explain the effectiveness of cyber threats and appropriate countermeasures. <i>TSO 5e.</i> Explain the fundamentals of Artificial Intelligence and AI agents. <i>TSO 5f.</i> Analyze types of learning paradigms and their applications. <i>TSO 5g.</i> Explain the concept of blockchain and distributed ledger.	Unit- 5.0: Emerging Technologies (8hrs) 5.1 Internet of Things (IoT): Concept and architecture, Applications (Smart homes, Smart cities) 5.2 Cyber Security: Basic security Terminologies, CIA Triad, Types of cyber threats and countermeasures 5.3 Artificial Intelligence (AI): Introduction, AI agents, Applications. 5.4 Machine Learning (ML): Types of learning paradigms, Real-world applications. 5.5 Blockchain: Concept of distributed ledger, Blocks, hashing, decentralization. 5.6 Data Science: Data Lifecycle, Role of data in decision making.

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	<i>TSO 5h.</i> Analyze the role of hashing and decentralization in blockchain systems. <i>TSO 5i.</i> Describe the concept of data science and the stages of the data science lifecycle. <i>TSO 5j.</i> Analyze the role of data in decision making.	

Reference/Books

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
1.	Computer Fundamentals	P. K. Sinha and Priti Sinha	BPB Publications, 6 th Edition, ISBN: 978-8176567527
2.	Introduction to Computers	Peter Norton	McGraw Hill, 7 th Edition, ISBN: 978-0070636774
3.	Data Representation and Computer Logic	Charles M. Gilmore	McGraw Hill, ISBN: 978-0070245754
4.	Operating System Concepts	Abraham Silberschatz, Peter B. Galvin and Greg Gagne	Wiley, 10 th Edition, ISBN: 978-1119456339
5.	Data Communications and Networking	Behrouz A. Forouzan	McGraw Hill, 5 th Edition, ISBN: 978-0073376226
6.	IoT for Beginners: Architecture, Principles and Applications	Amita Kapoor, Vinay Chopra	BPB Publications, 1 st Edition, ISBN: 978-9355510063
7.	Fundamentals of Cyber Security	Mayank Bhushan, Rajkumar Singh Rathore, Aatif Jamshed	BPB Publications, 1 st Edition, ISBN: 9789386551559
8.	Artificial Intelligence: A Guide for Thinking Humans	Melanie Mitchell	Farrar, Straus and Giroux, 1 st Edition, ISBN: 9780374257835